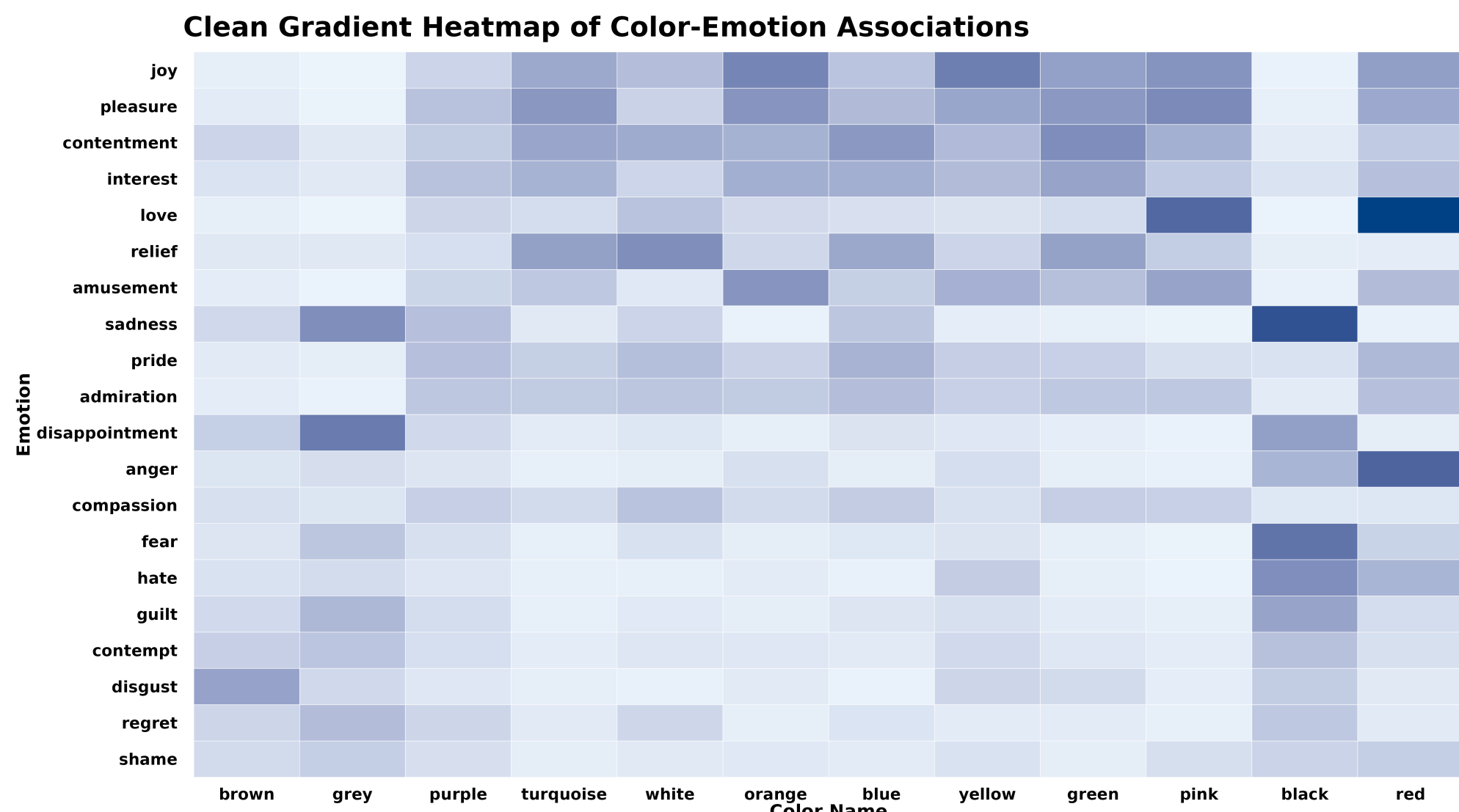


The Universal Language of Colors

The Psychology of Color: Universal or Cultural?

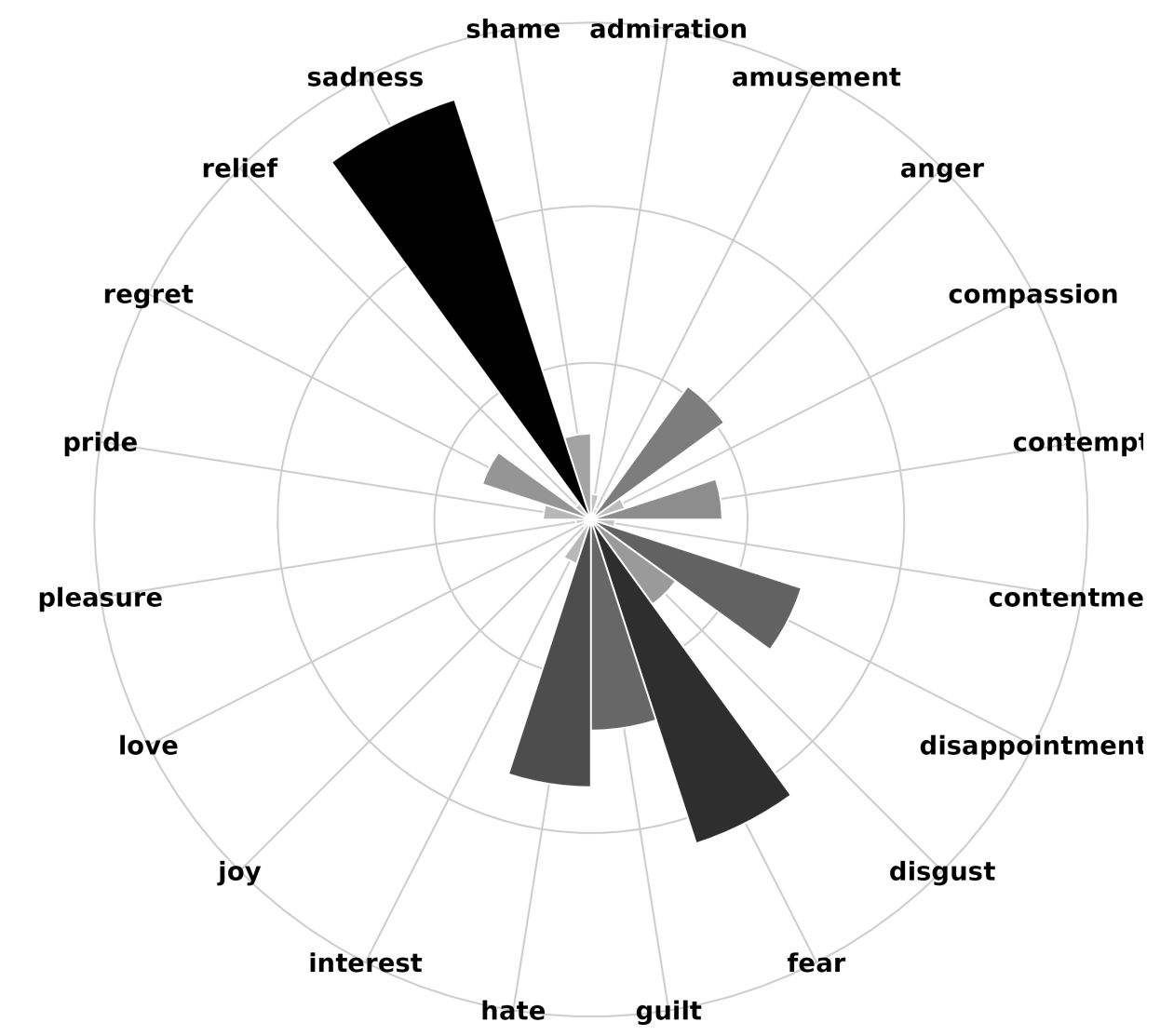
Colors are not merely visual stimuli; they are powerful emotional components that shape how we perceive the world around us. This study addresses a long-standing question in psychology: Are emotional responses to colors universal and biologically driven, or are they shaped by cultural context? To explore this question, the project analyzes a reliable dataset published by Jonauskaitė et al. (2019) in *Royal Society Open Science*. The dataset includes 8,532 observations from 711 participants across four regions: China (representing Eastern culture) and Germany, Greece, and the United Kingdom (representing Western culture). By examining these cultural contrasts, this study aims to determine whether color-emotion associations reflect a universal language or a culture-specific experience.



The heatmap reveals clear patterns in color-emotion associations, with negative emotions clustering around black, gray, and brown, and positive emotions around red, yellow, orange, and pink. Notably, red evokes both highly positive and highly negative emotions, while blue and green show more balanced emotional profiles.

The Emotional Depth of Black

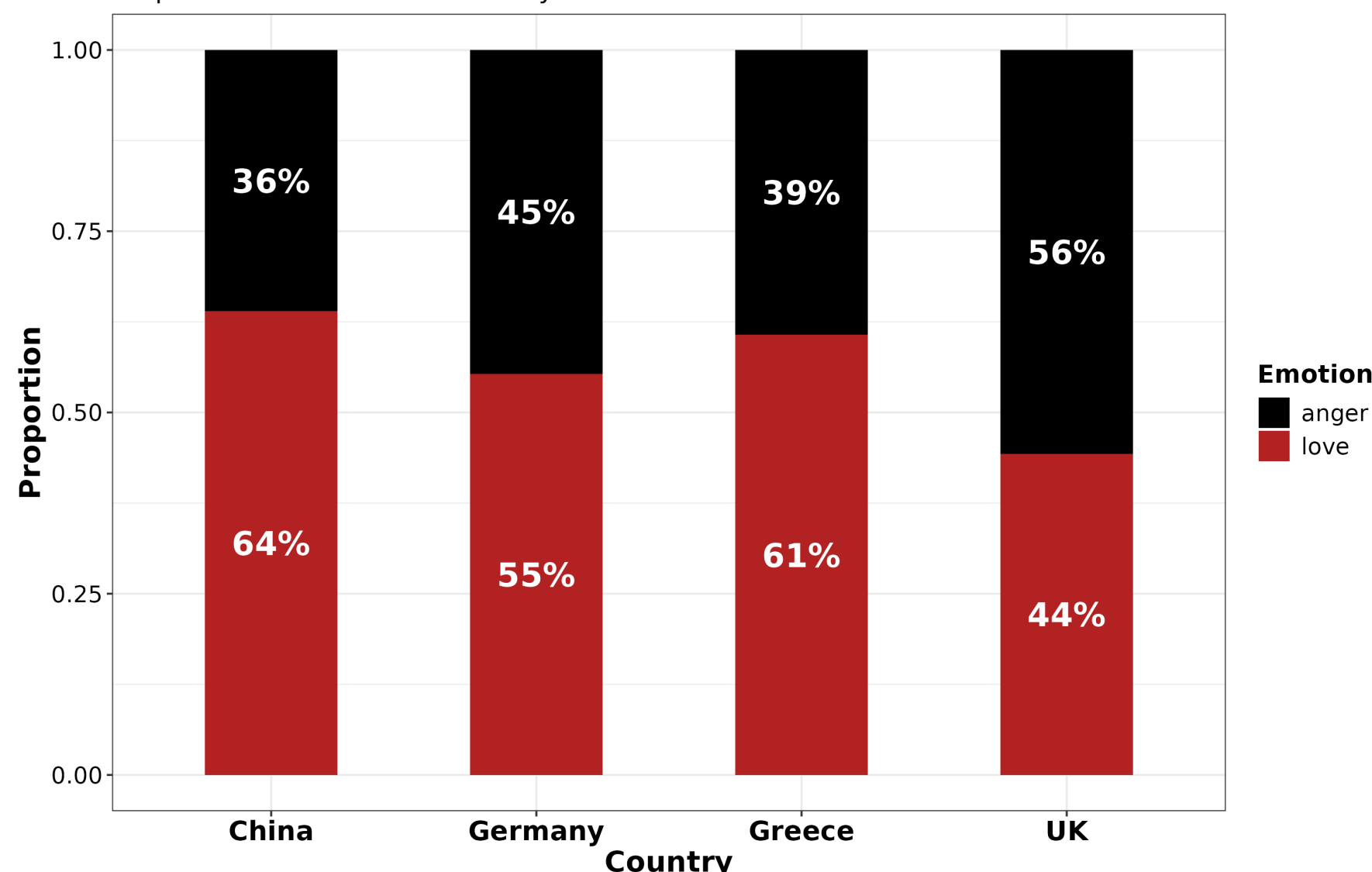
Fear, Sadness, and Hate: A distinct emotional signature



The radar chart shows that the color black is strongly and consistently associated across cultures with negative emotions, particularly sadness, followed by fear, hate, and guilt. In contrast, black exhibits little to no association with positive emotions such as joy and love.

The Dual Nature of Red: Love and Anger

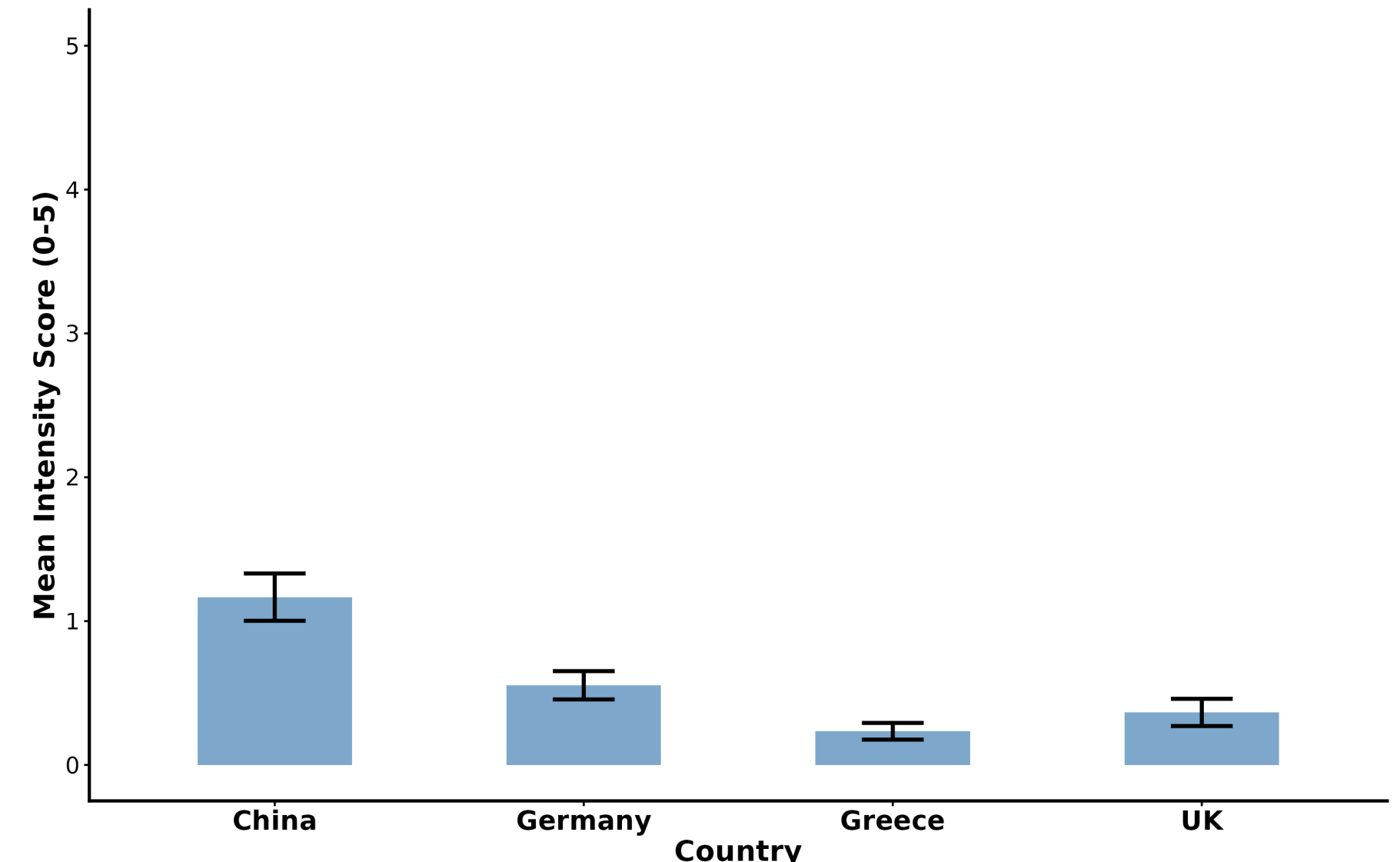
Proportion of emotional intensity for red across cultures



The bar chart highlights the dual emotional nature of the color red, which is associated with both love and anger across cultures. While red is predominantly linked to love in China, Greece, and Germany, the United Kingdom stands out as the only case where anger surpasses love. This pattern suggests that although the love-anger duality of red is widely shared, the dominant emotional response varies across cultural contexts.

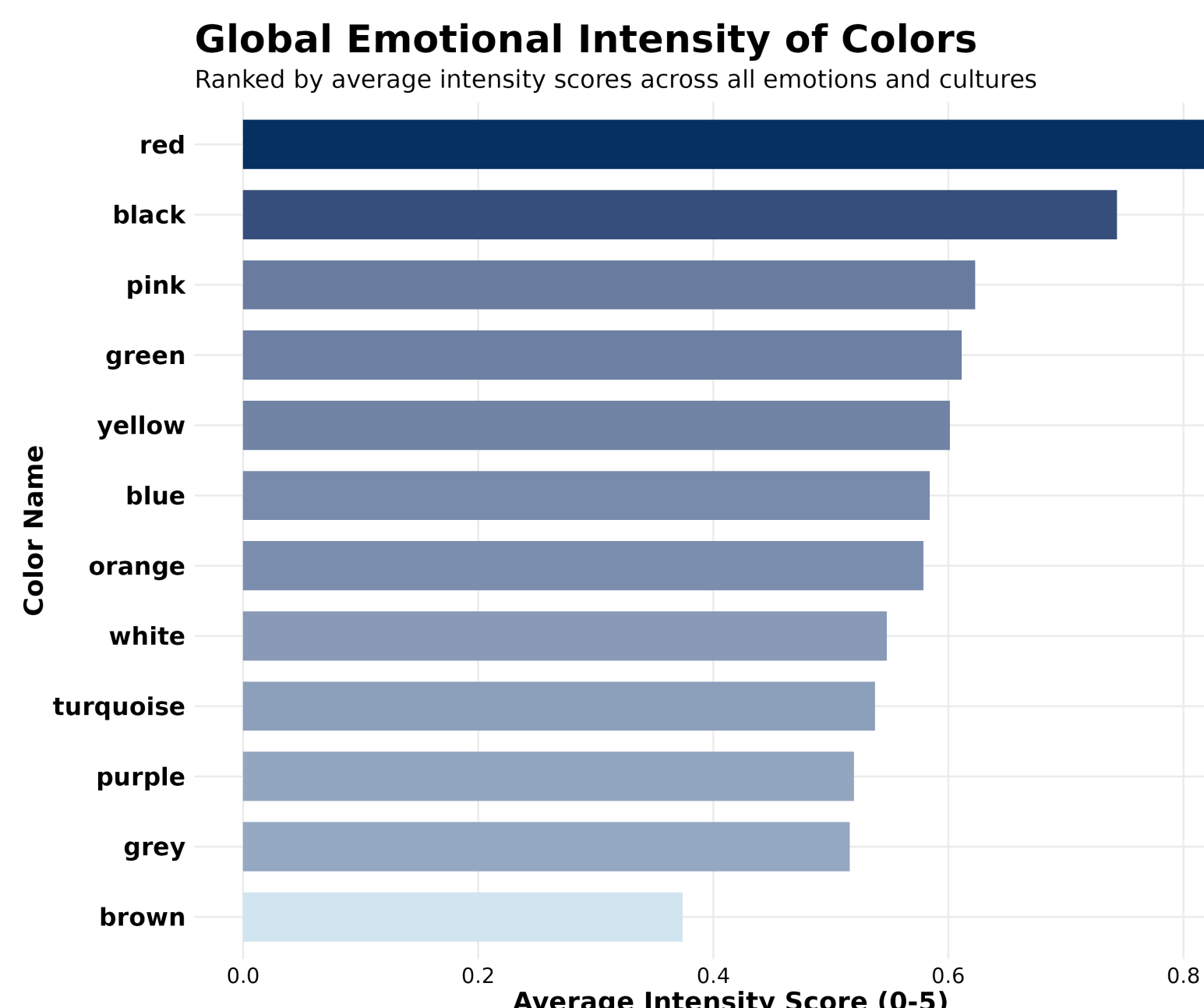
Sadness Intensity for White: Cultural Comparison

Bars show the mean score, and error bars show the Standard Error



The association between the color white and sadness varies markedly depending on cultural context. While this association is weak in Western countries, the strong linkage between white and sadness in China clearly highlights the decisive role of cultural background.

The horizontal bar chart ranks colors based on their average emotional intensity scores across all 20 emotions and 4 cultures. Red emerges as the most emotionally charged color globally, reflecting its powerful ability to trigger intense reactions in both positive (love) and negative (anger) spectrums. Black follows closely as the second most intense color, primarily driven by its strong and universal associations with heavy negative emotions. In contrast, brown is identified as the least emotionally evocative color in the dataset, showing significantly lower intensity scores compared to the rest of the palette. These rankings provide a definitive hierarchy of how "loudly" different colors speak to the human psyche, regardless of cultural background.



Findings reveal that color-emotion associations encompass both cross-cultural commonalities and culture-specific variations. Red and black trigger the highest levels of global psychological arousal and are linked to the most intense emotional responses. Cultural context remains decisive; for instance, while white is predominantly perceived as neutral in Western cultures, it is strongly associated with sadness in China. Similarly, red is primarily linked to love in China, whereas it is more frequently associated with anger in the United Kingdom.

